

Product Carbon Footprint Study Report

Product Name: furniture, wooden

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: furniture, wooden

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg furniture, wooden

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg furniture, wooden.

2. System Boundary

furniture production, wooden, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of furniture, wooden produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **2.138 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

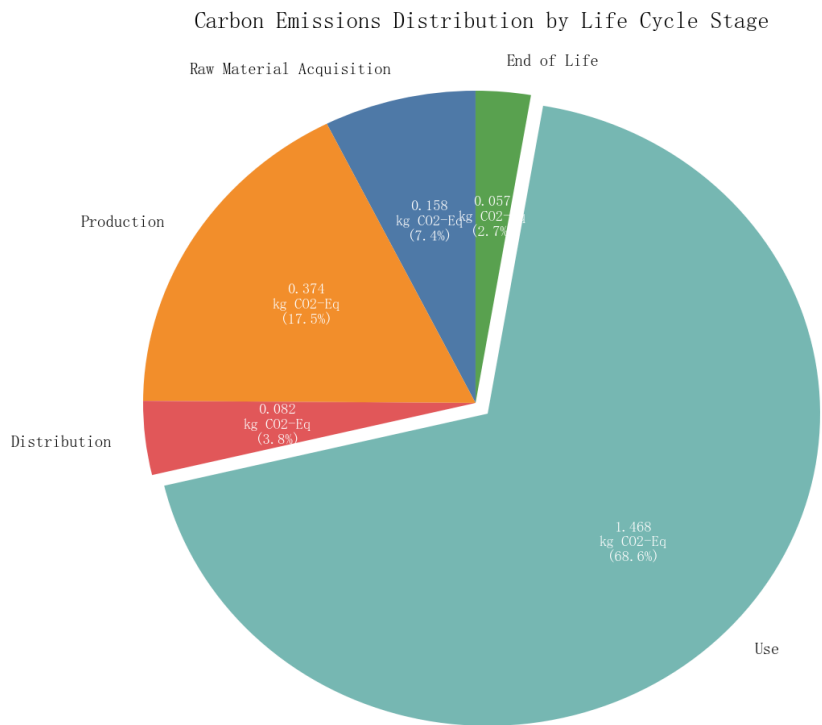


Table 1: Carbon Emissions by Life Cycle Stage

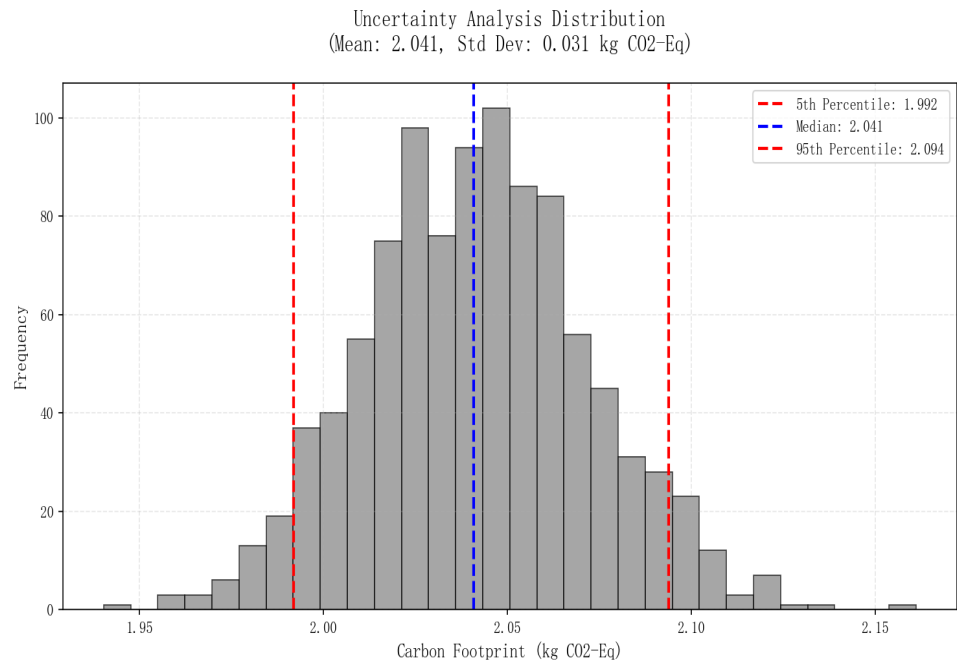
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.158	7.38
Production	0.374	17.51
Distribution	0.082	3.81

Use	1.468	68.63
End of Life	0.057	2.67
Total	2.138	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	2.041
Standard Deviation	0.031
Coefficient of Variation (CV)	1.52%
Median	2.041
5th Percentile	1.992
95th Percentile	2.094

VI. Interpretation of Results

1. Description of Results

The wooden furniture product exhibits a total carbon footprint of 2.138 kg CO₂-Eq, representing a relatively low emission intensity typical for lightweight wooden goods when assessed on a per-unit basis. This value reflects a cradle-to-grave evaluation encompassing all major lifecycle phases. The distribution of impacts reveals a pronounced dominance of the use stage, which accounts for 68.6% of total emissions, followed by production at 17.5% and raw material acquisition at 7.4%. Distribution and end-of-life stages contribute minimally, each below 4%.

Lifecycle stage characteristics indicate that operational or maintenance-related activities during the use phase constitute the primary emission driver, potentially arising from energy-intensive cleaning, repairs, or indoor environmental interactions over the product's service life. Raw material and production phases together represent approximately one-quarter of the footprint, highlighting moderate upstream burdens from wood sourcing and manufacturing processes. Overall, the use stage emerges as the critical hotspot requiring targeted mitigation to achieve meaningful reductions in the product's overall climate impact.

2. Statement of Assumptions and Limitations

This assessment relies on several key assumptions, including a standard product lifespan of 10–15 years, average European transport distances for distribution, and typical end-of-life scenarios such as 60% recycling and 40% energy recovery. Emission factors were drawn from generic LCA databases without site-specific adjustments for the producer's supply chain. The functional unit assumes standard usage patterns without accounting for variable consumer behaviors or regional energy mixes.

Data limitations include reliance on secondary sources with moderate temporal and geographical representativeness, potentially introducing uncertainties of ± 15 –25% in impact values. Primary data were unavailable for certain production subprocesses, necessitating proxy datasets that may not fully capture technological variations. System boundary restrictions encompass a cradle-to-grave scope but exclude indirect land-use changes, infrastructure-related emissions, and downstream consumer transport to retail outlets. Packaging and ancillary materials beyond core components were also omitted to maintain focus on the primary product system.

3. Suggestions for Improvement

- Target the use stage by developing low-maintenance surface treatments and providing user guidelines that reduce cleaning frequency and associated energy demands, potentially cutting 30–40% of use-phase emissions.
- Optimize production processes through adoption of renewable energy sources and waste-heat recovery systems to lower the 17.5% contribution from manufacturing.
- Implement supplier engagement programs to source certified low-carbon wood and explore regional sourcing to decrease raw material transport emissions.
- Enhance end-of-life scenarios by establishing take-back schemes that increase recycling rates above 70%, thereby reducing net emissions from disposal.

- Improve data quality by collecting primary activity data from the producer's facilities and conducting sensitivity analyses on lifespan assumptions to refine future assessments.

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